Enrollment No.: _____

Darshan UNIVERSITY

योगः कर्मसु कौशलम्

Darshan Institute of Computer Application

MCA | Sem-4 | Summer-2024

Course Code	:	2105CS412	**Date**	:	10-04-2024
Course Name	:	Computer Networks	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:

1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1 **(A)** What is topology? Explain star topology in brief. **4**

(B) Give difference between LAN, MAN and WAN. **3**

OR

What is the difference between a host and an end system? List several different types of end systems.

(C) Discuss the five layer internet protocol stack along with the functionalities of each layer in detail. **7**

OR

Discuss the circuit switching versus packet switching approaches for moving data through a network of links and switches.

Q.2 **(A)** Explain following terms: 1) Propagation Delay 2) Transmission Delay **4**

(B) Give difference between Hub and Repeater. **3**

OR

Give difference between DSL and ADSL technologies.

(C) Discuss different types of guided and unguided media used to transmit data in network. **7**

OR

Explain Electromagnetic spectrum in detail.

Q.3 **(A)** Explain paritybit check with example. **4**

(B) Give differences between Connection oriented versus Connection less services. **3**

OR

List role & responsibility in datalink layer.

(C) Explain checksum technique with example. **7**

OR

A bit stream 1101011011 is transmitted using standard CRC method. The generator polynomial is x^4+x+1. Calculate CRC code.

Q.4	**(A)**	Discuss the count-to-infinity problem in distance vector routing algorithm with example.	**4**
	(B)	How many subnets can be created for the subnet mask 255.255.255.224? Which IP address class this subnet does belong to?	**3**
		OR	
		List the classes in classful addressing.	
	(C)	Explain IPv4 datagram format and importance of each filed.	**7**
		OR	
		Explain Link State Routing algorithm in detail.	
Q.5	**(A)**	Draw FSM diagram for rdt2.0 protocol.	**4**
	(B)	Differentiate HTTP non-persistent and persistent connection.	**3**
		OR	
		Differentiate HTTP GET method and POST method.	
	(C)	How pipeline approach improves the overall sender utilization time? Explain Go-Back-N pipeline approach in transport layer.	**7**
		OR	
		Explain TCP Congestion mechanism in detail.	

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